

Authentication & User Management

1. User Registration

- Design **registration form UI**
- Allow role selection (**student / teacher**)
- Implement **frontend form validation**
- Send registration data to backend API
- Check **email uniqueness**
- **Hash password using bcrypt**
- Store user data in database
- Return **JWT authentication token**

2. User Login & Authentication

- Create **login page**
- Validate login credentials
- Verify password with bcrypt
- Generate **JWT token**
- Store token on frontend
- Protect routes that require authentication
- Implement **logout functionality**

3. User Profile Management

- Create **profile page UI**
- Fetch user data from API
- Display profile information
- Allow editing of:
 - Name
 - Contact number
 - Profile picture

- Update user information in database

4. Password Management

- Allow users to **change password**
- Validate new password strength
- Confirm password before update
- Hash updated password before saving

5. Account Deletion

- Add **delete account option**
- Display confirmation prompt
- Remove user from database
- Handle related data (optional cleanup)

6. Role-Based User Handling

- Store **user role (student / teacher)**
- Control access to features based on role
- Ensure system recognizes user type during login

7. Input Validation & Security

- Implement **frontend validation**
- Implement **backend validation**
- Sanitize user inputs
- Prevent duplicate emails
- Protect API routes with authentication middleware

8. Database Structure

- Create **User schema (Mongoose)** including:
 - Name
 - Email
 - Password

- Role
- Contact details
- Profile picture
- Created date
- Apply schema validations

Common User Profile Details (Both Student & Teacher)

- User ID
- Name
- Email
- Password (hashed)
- Role (Student / Teacher)
- Contact Number
- Profile Picture
- Account Created Date

Student Profile Details

- Student ID
- Date of Birth
- Gender
- Grade
- School / Institution Name
- Subjects Interested In – according to grade
- Teacher Profile Details
- Teacher ID
- Qualification / Degree
- Teaching Subjects
- Years of Experience
- Institution / Organization (optional)
- Availability Schedule

